18.A Chemical Thermodynamics I

Spontaneous Processes. Of the apparent infinity of chemical equations that we could scribble down on a piece of paper, only a very small subset "work as advertised." It is entirely possible that, when chemical substances are mixed, they do not engage with each other at all and maintain their chemical identities. On the other hand, some combinations of chemicals lead to violent and complete chemical reactions without any outside intervention—they happen "spontaneously." A major goal of this chapter is to develop a criterion that enables us to determine whether a given process will happen spontaneously or not. We start by defining exactly what we mean by a "spontaneous process."

- Much of the work of chemists is to make predictions and answer questions about physical and chemical processes.
    - *Reactions and Stoichiometry:* What happens and in what proportions?
    - *Kinetics:* At what speed does a reaction occur?
    - *Equilibrium and Thermodynamics:* To what extent does a reaction occur, given infinite time?
- The basic question of this chapter is whether a physical or chemical process will or will not occur spontaneously starting from a standard set of initial conditions.
- A **spontaneous process** occurs without continuing outside influence—it "just goes."
- Spontaneity and time are theoretically *unrelated*. Spontaneous processes can be slow!
    - If A → B occurs spontaneously, the reverse process B → A is physically impossible (**non-spontaneous**). See the GIFs below (click for moving images) or this page.


ice cubes → water
spontaneous


water ↛ ice cubes
non-spontaneous

- What do all spontaneous processes have in common? This is fundamentally a question about the "arrow of time." In what "direction" does it point? How do we conceptualize and quantify that direction?

- Ideally, we'd like a state function that tells us whether a given process is spontaneous or not based on the sign of the state function (or a change therein). Consider gravitational potential energy as a metaphor from physics.

- **Entropy (*S*)** is a measure of disorder, randomness, or (best) matter and energy dispersal in a system. It is the state function we need! Consider popping a balloon…

- **The second law of thermodynamics:** spontaneous processes are associated with an increase (or no change) in the entropy of the universe S_{univ}.

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} \geq 0$$

The Classical Definition of Entropy; Units of Entropy. Entropy grew out of the study of heat engines, the purpose of which is to convert a temperature difference into mechanical work. Different systems transfer different amounts of heat even at the same temperature; entropy was defined as the ratio of heat to temperature under specific conditions. Although we won't use the classical definition of entropy for any calculations, it's very important for conceptual understanding. Take some time to meditate on entropy!

- Classical thermodynamics defines infinitesimal entropy change dS as the infinitesimal heat transferred to or from the system reversibly (δq_{rev}) divided by the temperature T at which this heat transfer occurs,

$$dS = \frac{\delta q_{rev}}{T}$$

- The units of entropy are thus **energy per temperature**, most commonly Joules per Kelvin.

- Add heat and the entropy of the system increases;
  remove heat and the entropy of the system decreases.

- Conceptually, entropy is a measure of the *amount of dispersed energy in the system* divided by the *temperature of the system.*

    - Systems with a greater diversity of "ways to store" energy at a given temperature, such as bigger and "wigglier" molecules, have higher entropy.

    - Further highly instructive reading: https://franklambert.net/entropysite.com/#summary.

Factors Affecting Entropy. The entropy change for a chemical reaction is needed to determine the conditions under which it will be spontaneous (if any…!). With an understanding of structural factors affecting the entropy of chemical substances, we can make an educated guess concerning the sign of entropy change. We'll soon see that this gets us well on our way to predicting spontaneity.

- Factors affecting the entropy of a system (S) or entropy change of a process (ΔS)…

    - Phase of matter

    - Temperature

    - Volume of a gas

    - Amount (in moles) of a gas

    - Amount (in moles) of distinct types of compounds

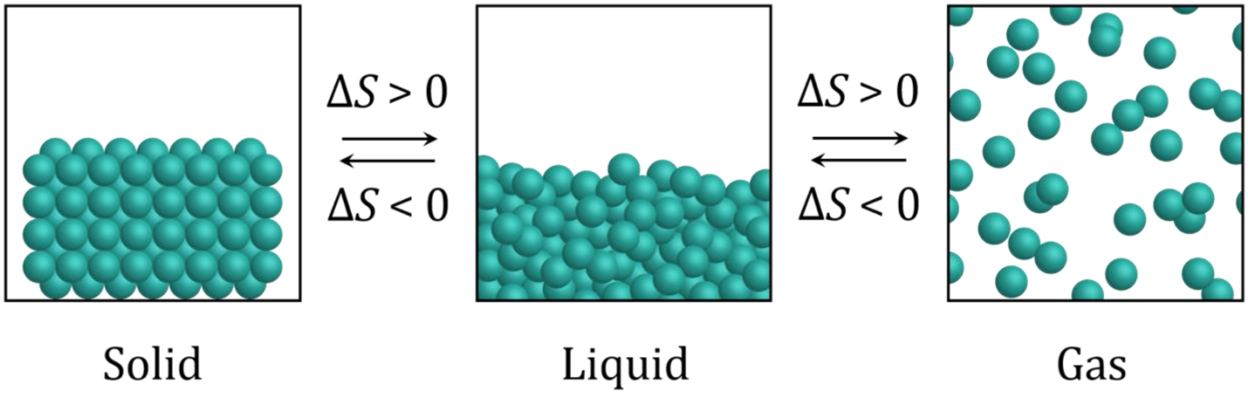


Figure 18.1. Entropy changes for phase transitions involving solid, liquid, and gas.

For which process(es) would you expect the change in entropy to be less than zero? Select all that apply.

A. $I_2(s) \rightarrow I_2(g)$

B. $H_2O(s) \rightarrow H_2O(l)$

C. $Ca_3(PO_4)_2(s) \rightarrow 3\ Ca^{2+}(aq) + 2\ PO_4^{3-}(aq)$

D. $2\ O_2(g) + 2\ SO(g) \rightarrow 2\ SO_3(g)$

E. Condensation

For which process would you expect ΔS^{o} to be *most* positive?

A. $O_2\,(g) + 2\ H_2(g) \rightarrow 2\ H_2O(g)$

B. $NH_3(g) + HCl(g) \rightarrow NH_4Cl(g)$

C. $2\ NH_4NO_3(s) \rightarrow 2\ N_2(g) + O_2(g) + 4\ H_2O(g)$

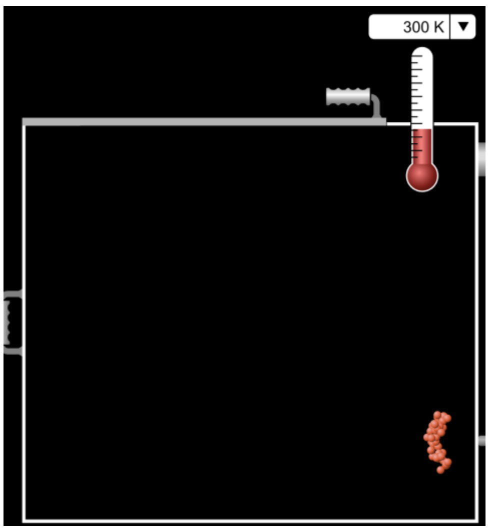
D. $N_2O_4(g) \rightarrow 2\ NO_2(g)$

E. $H_2O(l) \rightarrow H_2O(s)$

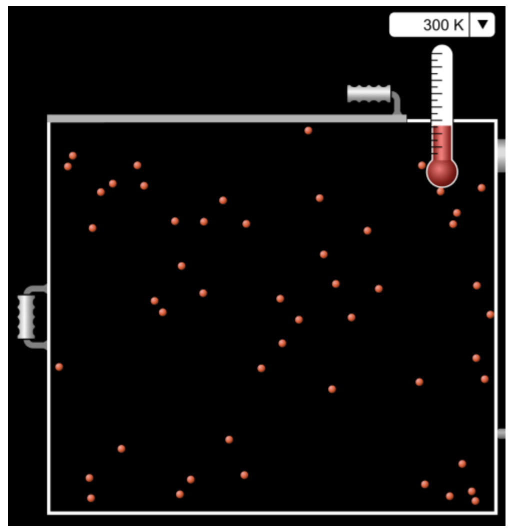
Entropy and Probability. So far, we've focused on entropy as a measure of the extent of energy dispersed in a system at a given temperature. An equivalent formulation of entropy, due initially to Boltzmann, is based on the probability of a system existing in its current state on the macroscopic scale. Boltzmann realized that "the entropy of the universe tends toward a maximum" ($\Delta S_{univ} \geq 0$) is equivalent to the tautological "thermodynamic systems tend spontaneously toward their maximally likely macrostate." We'll explore the latter formulation of the second law of thermodynamics here and provide a statistical definition for entropy.

- Consider the following two "snapshots" of an ideal gas at given T and n.
  Which feels more likely to observe? What is the direction of the "arrow of time" here?

state 1



state 2



- Of course, the arrow of time points from state 1 to state 2. The gas expands spontaneously into the empty space in the container.
  - The effective volume of the gas increases from state 1 to state 2. $\Delta V > 0$.
  - The kinetic energy of the gas particles and the particles themselves disperse throughout the container in the process. Thus, $S_2 - S_1 = \Delta S > 0$.
  - State 1 will almost never arise if the system starts in state 2.
- The system evolves spontaneously from state 1 to state 2 because state 2 is exponentially more likely to exist than state 1.

- The likelihood W of a **macrostate** is equal to the number of **microstates** that give rise to that macrostate.
    - The **macrostate** of a system is what we've just called the "state" so far; it is defined by macroscopically measurable variables such as V, T, and n. The macrostate is stable for a system in equilibrium.
    - A **microstate** is a specific set of submicroscopic particle positions and velocities $\{\boldsymbol{r}_i, \boldsymbol{v}_i\}$. Macrostate is an average over all accessible microstates. The microstate is always changing, even for systems in equilibrium.
- Higher W is associated with all the structural factors we've already identified that contribute to higher S: larger volume, higher temperature, greater mass, more "wiggly" molecules, and more gaseous substances.
- The logarithm of W is proportional to the entropy S. The proportionality constant between the two is **Boltzmann's constant $\boldsymbol{k_B}$** (1.381×10^{-23} J/K).

$$S = k_B \ln W$$

- Consider a system containing N gas particles (equivalently, n moles times Avogadro's number N_A) occupying a volume V at temperature T. The likelihood W of this state varies exponentially with the number of particles.

$$W \propto V^{n \cdot N_A}$$

$$S \propto k_B \ln V^{n \cdot N_A} = nN_A k_B \ln V = nR \ln V$$

- The gas constant is Boltzmann's constant. Boltzmann's constant is the gas constant. One is at the scale of 1 mole (R) and the other is at the scale of 1 molecule (k_B). Conceptually, they are the same!
- For an ideal gas undergoing a change in volume from V_1 to V_2,

$$\Delta S = nR \ln V_2 - nR \ln V_1 = nR \ln \left(\frac{V_2}{V_1} \right)$$

The Third Law and Standard Molar Entropies. Reflecting on Boltzmann's entropy equation, we can at least *conceive* of a system for which $W = 1$ and $S = 0$. It follows that there is an absolute zero of entropy, associated with a (hypothetical) system at absolute zero with a single accessible microstate (a "perfect crystal"). At this point, chemists have achieved temperatures extraordinarily close to absolute zero and literally watched entropy approach zero in the process. From these observations, the third law of thermodynamics has emerged.

- The **third law of thermodynamics** establishes an absolute zero of entropy: the entropy of a perfectly crystalline material at zero Kelvin.
    - At absolute zero, all atoms or molecules are in the lowest-energy state possible.
    - All molecules are in identical orientations and environments.

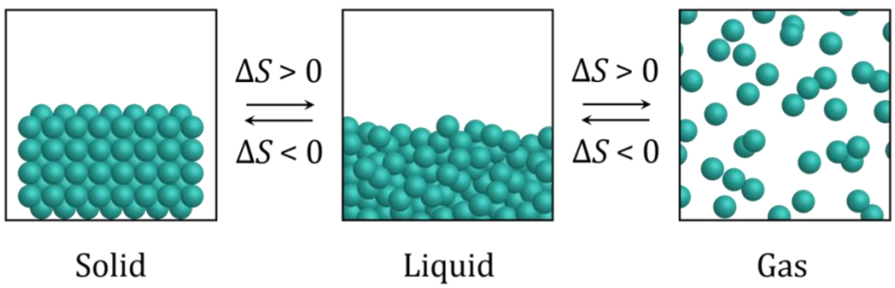


Figure 18.1. Entropy changes for phase transitions involving solid, liquid, and gas.

- **Standard molar entropy ($S°$)** is an absolute entropy value for 1 mole of a substance at a given temperature based on the zero point established by the third law.
    - Units are Joules per Kelvin per mole ($J{\cdot}K^{-1}{\cdot}mol^{-1}$).
    - Recall that the ° symbol indicates that the substance is under standard conditions: 1.00 atm for gases and 1.00 mol/L for solutes.
- What structural factors affect the entropy of a substance? We already know!

Compound	$S°$ ($J\ K^{-1}\ mol^{-1}$)	Compound	$S°$ ($J\ K^{-1}\ mol^{-1}$)
$CO_2(g)$	213.8	$HCl(g)$	186.9
$CaO(s)$	38.1	$Hg(l)$	75.9
$CaCO_3(s)$	91.7	$Hg(g)$	175.0
$H_2(g)$	130.7	$I_2(s)$	116.1
$H_2O(l)$	70.0	$I_2(g)$	260.7
$H_2O(g)$	188.8	$O_2(g)$	205.2

Table 18.1. Standard molar entropies for select substances.

- **Standard entropy of reaction ($\Delta S°$)** can be calculated from standard molar entropies. This is just an application of entropy as a state function and Hess's law. For our general reaction with *M* reactants and *N* products, it's "products minus reactants" again.

$$r_1\,\mathrm{R}_1 + r_2\,\mathrm{R}_2 + \ldots + r_M\,\mathrm{R}_M \rightarrow p_1\,\mathrm{P}_1 + p_2\,\mathrm{P}_2 + \ldots + p_N\,\mathrm{P}_N$$

$$\Delta S^{\circ}_{rxn} = \sum_{j}^{N} p_j S^{\circ}(\mathrm{P}_j) - \sum_{i}^{M} r_i S^{\circ}(\mathrm{R}_i)$$

- There is no need to consider formation reactions here, since there is an absolute zero of entropy and standard entropies are measured with respect to it.

Which substance would you expect to have the *greatest* standard molar entropy?

A. $CH_3OH(l)$

B. $CH_3OH(g)$

C. $CH_3CH_2OH(l)$

D. $CH_3CH_2OH(g)$

E. $H_2O(g)$

18.B Chemical Thermodynamics II

The Second Law Revisited; Gibbs Free Energy. The second law tells us that all physically possible processes increase the entropy of the universe (or leave it unchanged). Because the universe is unfathomably large, this statement of the second law is too general to be useful. As chemists, we would like to somehow "reformulate" $\Delta S_{univ} \geq 0$ into an equation that focuses on the system of interest. We can do that by placing some constraints on how processes happen; when we do, a new state function emerges that we can use directly to predict the spontaneity of a process. We're finally about to arrive at the thermodynamic analogue of gravitational potential energy or height that we were seeking at the start of the chapter: a state function that always drops in value for spontaneous processes.

- According to the second law, ΔS_{univ} is positive or zero for any spontaneous process.
- Because $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$, this can happen when either or both ΔS_{sys} and ΔS_{surr} is/are positive…
  - $\Delta S_{sys} > 0$, $\Delta S_{surr} > 0$ → spontaneous
  - $\Delta S_{sys} < 0$, $\Delta S_{surr} < 0$ → non-spontaneous
  - $\Delta S_{sys} < 0$, $\Delta S_{surr} > 0$ → spontaneous when $|\Delta S_{surr}| > |\Delta S_{sys}|$
  - $\Delta S_{sys} > 0$, $\Delta S_{surr} < 0$ → spontaneous when $|\Delta S_{sys}| > |\Delta S_{surr}|$
- Exothermic processes ($\Delta H_{sys} < 0$) are more likely than endothermic process ($\Delta H_{sys} > 0$) to be spontaneous…but some endothermic processes are spontaneous.
- Processes with a positive entropy change in the system ($\Delta S_{sys} > 0$) are more likely to be spontaneous than processes with a negative entropy change in the system ($\Delta S_{sys} < 0$) …but there are some spontaneous processes with $\Delta S_{sys} < 0$.
- To make sense of this, we need to reformulate the second law to (a) incorporate enthalpy and (b) focus on the system. We start with a spontaneous process occurring at constant *T* and *P*.

$$\Delta S_{sys} + \Delta S_{surr} \geq 0$$

- The entropy change in the surroundings is equal to the heat absorbed by the surroundings divided by the temperature *T*.
- The heat absorbed by the surroundings at constant *P* is $-\Delta H_{sys}$!

$$\Delta S_{surr} = \frac{q_{surr}}{T} = -\frac{\Delta H_{sys}}{T}$$

$$\Delta S_{sys} - \frac{\Delta H_{sys}}{T} \geq 0$$

- Now, we multiply both sides by –*T* to reveal a change in a new state function that must be negative for this (given) spontaneous process. The state function *G* is called **Gibbs free energy** or **chemical potential**.[1]

$$\Delta H_{sys} - T\Delta S_{sys} \leq 0$$

$$G = H - TS$$

$$\Delta G_{sys} \leq 0$$

- Gibbs free energy is the enthalpy of the system minus the product of temperature and entropy. "Free" here means "available."
    - *H* is the internal energy *U* plus *PV*, the total energy in the system that be transferred into work based on the first law.
    - *TS* is the quantity of energy in the system that is randomly distributed and thus unavailable to do useful work in accordance with the second law: the "second-law tax."
    - For a spontaneous process, Δ*G* is the maximum useful work (w_{max}) that the process can "do" on its surroundings.
    - For a non-spontaneous process, Δ*G* is the minimum work that must be invested to "make the process go."
- For a process occurring under standard conditions (1.00 atm pressure for all gases and 1.00 mol/L for all solutes),

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

[1] The term "chemical potential" is meant to evoke the idea of potential energy. *G* is a kind of abstract "height" of a chemical substance in "energy space." The analogy runs deep!

- **Standard free energy of formation (ΔG_f°)** is the free energy change for the formation reaction of a compound occurring under standard conditions.

- Two ways to calculate the **standard free energy of reaction (ΔG°_{rxn})**,

$$r_1\,\mathrm{R}_1 + r_2\,\mathrm{R}_2 + \ldots + r_M\,\mathrm{R}_M \rightarrow p_1\,\mathrm{P}_1 + p_2\,\mathrm{P}_2 + \ldots + p_N\,\mathrm{P}_N$$

  - Using tabulated free energies of formation at a given temperature

$$\Delta G^\circ_{rxn} = \sum_j^N p_j \Delta G^\circ_f(\mathrm{P}_j) - \sum_i^M r_i \Delta G^\circ_f(\mathrm{R}_i)$$

  - Using standard enthalpies of formation and standard entropies—we've already seen both!

$$\Delta G^\circ_{rxn} = \Delta H^\circ_{rxn} - T\Delta S^\circ_{rxn}$$

Spontaneity (Sometimes) Depends on Temperature. Because temperature appears in the definition of Gibbs free energy, we should expect spontaneity to depend on temperature in general. Here, we'll come to understand cases in which the spontaneity of a process does or does not depend on *T*. Some processes are physically impossible regardless of the temperature—recall the self-organizing ice cubes at the start of the chapter—and some are spontaneous regardless of the temperature. Still others are only spontaneous at low or high temperatures.

- The value and sign of ΔG depend on temperature.

ΔH	ΔS	ΔG	Reaction Characteristics
+	+	+ or –	Spontaneous at high temperatures Nonspontaneous at low temperatures
+	–	+	Nonspontaneous at all temperatures
–	+	–	Spontaneous at all temperatures
–	–	– or +	Spontaneous at low temperatures Nonspontaneous at high temperatures

- For processes with ΔH and ΔS having the same sign, the critical temperature T_{crit} at which the process becomes spontaneous is the temperature at which $\Delta G = 0$.

$$\Delta G = \Delta H - T_{crit}\Delta S = 0 \Rightarrow T_{crit} = \frac{\Delta H}{\Delta S}$$

What is the melting point of a compound A? The standard enthalpy and entropy of fusion for A are listed below. Enter your response in Kelvin to the nearest 1 K.

$$A(s) \rightarrow A(l)$$

ΔH^{o} = 8.1 kJ·mol^{-1}
ΔS^{o} = 42.4 J·K^{-1}·mol^{-1}

- Processes that are spontaneous only at low temperatures are called **enthalpy driven**. They are "encouraged" by exothermic ΔH.
- Processes that are spontaneous only at high temperatures are called **entropy driven**. They are "encouraged" by positive ΔS.

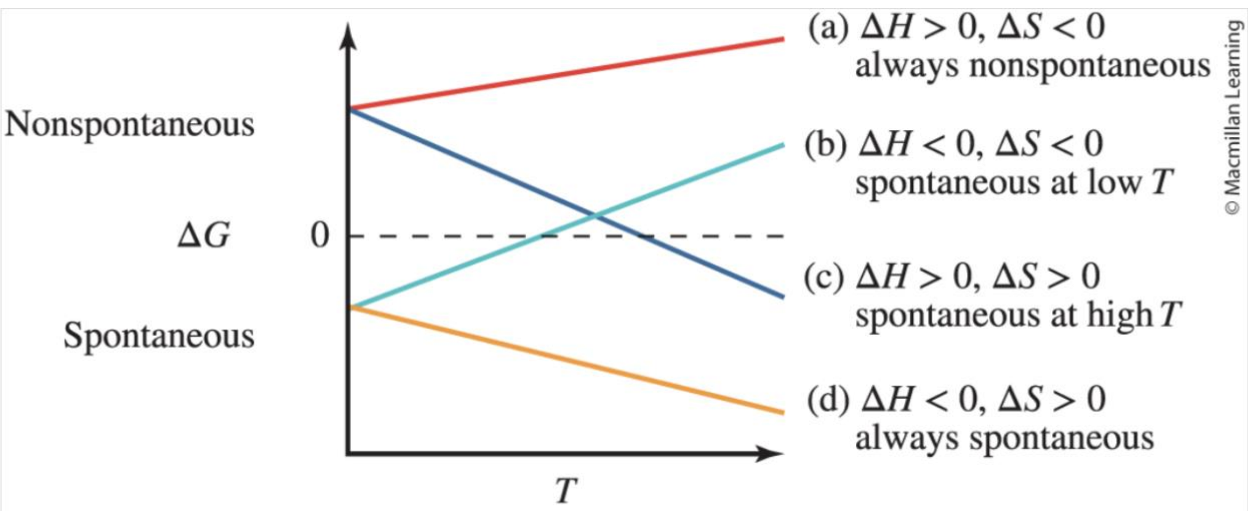


Figure 18.7. Graphical dependence of free energy change on temperature.

At constant pressure, the reaction below is endothermic.

$$N_2O_4(g) \rightarrow 2\ NO_2(g)$$

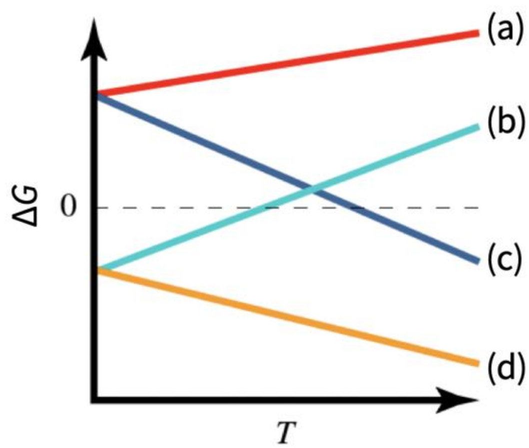
The reaction (as written) is…

A. Always spontaneous
B. Never spontaneous
C. Spontaneous only above a critical temperature
D. Spontaneous only below a critical temperature

Which line on the graph of Δ*G* versus *T* below corresponds to the hypothetical chemical reaction below?

$$A(g) + B(g) \rightarrow AB(g) + \text{heat}$$

A. Line (a) – red
B. Line (b) – cyan
C. Line (c) – blue
D. Line (d) – orange



Chapter Summary. Some of the amazing things we can now do with entropy, free energy, and chemical thermodynamics…

1. Define and describe spontaneous and non-spontaneous processes.

2. Articulate and apply the second law of thermodynamics as a statement about the entropy change of the universe.

3. Infer the relative entropies of chemical systems (or changes in entropy during chemical reactions) using information about state of matter, mass, and other factors.

4. Apply the statistical definition of entropy to isothermal expansion or compression of an ideal gas; conceptualize Boltzmann's constant and the gas constant as fundamental measures of entropy.

5. Articulate and apply the third law of thermodynamics, which establishes a zero of entropy.

6. Calculate standard entropies of reaction using standard molar entropies.

7. Express the second law in terms of free energy change in a chemical system.

8. Define Gibbs free energy and explain how it reflects the first and second laws of thermodynamics.

9. Calculate standard free energies of reaction using either free energies of formation or standard enthalpies and entropies.

10. Apply the temperature dependence of free energy to predict how temperature affects the spontaneity of a process.